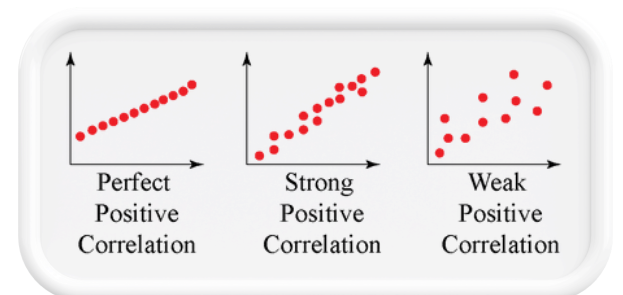
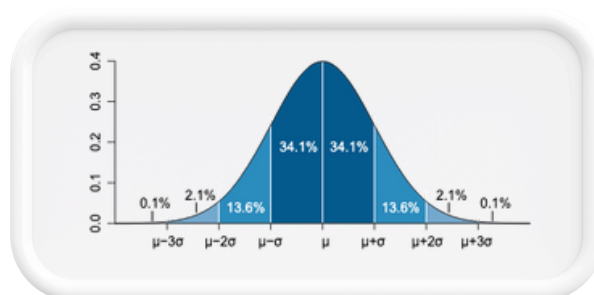
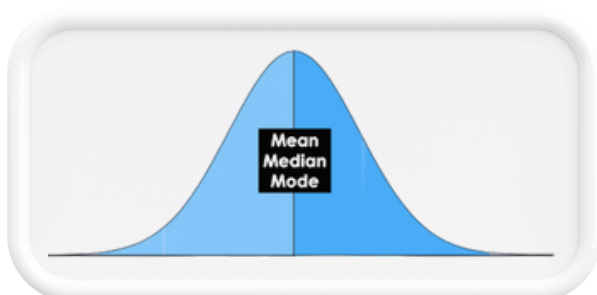


15 *Statistics Concepts* that you should know!



Mean, Median, and Mode

The classic trio.

- Mean is your average.
- Median is the middle value when data is sorted.
- Mode is the most frequent value.

Why it matters: These tell you different things about central tendency and they behave differently when outliers show up.

Standard Deviation and Variance

- Variance tells you how spread out your data is.
- Standard deviation is the square root of variance more interpretable, same idea.

Why it matters: You need to know how much your data varies. Otherwise, the average means nothing.

Probability Basics

This is the foundation of statistical thinking.

You need to understand:

- Independent vs. dependent events
- Conditional probability
- Bayes' Theorem (yes, at least intuitively)

Why it matters: Machine learning is built on probability. You can't dodge it.

Distributions

Not all data looks the same.

- Normal distribution is the bell curve.
- Binomial, Poisson, Exponential each has its use.

Why it matters: Different models assume different distributions. Get this wrong, and your results fall apart.

Outliers

An outlier is a data point that's way off from the rest. Could be a mistake. Could be gold.

Why it matters: Outliers can skew your stats. Always check them before trusting your mean.

Skewness and Kurtosis

- Skewness measures asymmetry.
- Kurtosis measures how heavy the tails are.

Why it matters: These give you a quick sense of shape and risk in your data.

Sampling Methods

How you collect data affects everything.

- Random sampling
- Stratified sampling
- Systematic sampling

Why it matters: Bad sampling = biased results. Period.

Central Limit Theorem

If you take a bunch of random samples, their means will tend to form a normal distribution even if the original data isn't normal.

Why it matters: This is why you can use normal-based inference on real-world, messy data.

Confidence Intervals

Not just a single value, but a range where your true value likely falls.

Why it matters: Helps communicate uncertainty the right way instead of pretending your number is perfect.

Hypothesis Testing

- **Null hypothesis:** no effect, no difference.
- **p-value:** how surprising your result is if the null were true.

Why it matters: It's the backbone of statistical inference. But don't treat $p = 0.049$ like gospel.

Correlation vs. Causation

Just because two things move together doesn't mean one causes the other.

Why it matters: You're here to solve real problems. Don't chase noise.

Correlation Coefficient

- Pearson: linear relationships.
- Spearman: ranked relationships.

Why it matters: Correlation $\neq$ causation, but it's a start.

ANOVA (Analysis of Variance)

Used when comparing more than two groups.

Why it matters: A t-test only works for two groups. ANOVA handles multiple groups in one go.

Regression Basics

Linear regression: models relationships.

Know your assumptions: linearity, homoscedasticity, independence.

Why it matters: This is your first predictive model. Learn it well.

Overfitting vs. Underfitting

Overfitting: too complex, memorizes noise.

Underfitting: too simple, misses patterns.

Why it matters: Every model lives somewhere on this spectrum. Your job is to find the sweet spot.